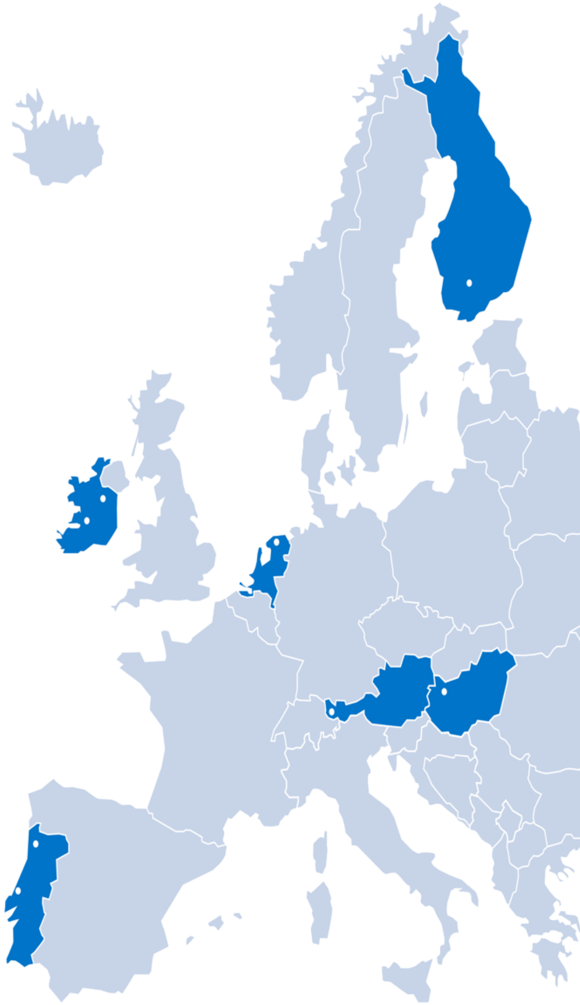

APPLIED CRYPTOLOGY: BASICS

BIP HACK YOUR DEVICE, 2026

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    - RSA, Diffie-Hellman, Elliptic Curves
  - Hashfunctions and Signatures; MAC
  - Certificates
  - Public Key Infrastructure (PKI)

  - Cracking Passwords and Hashes

Do you have any prior knowledge on cryptology?

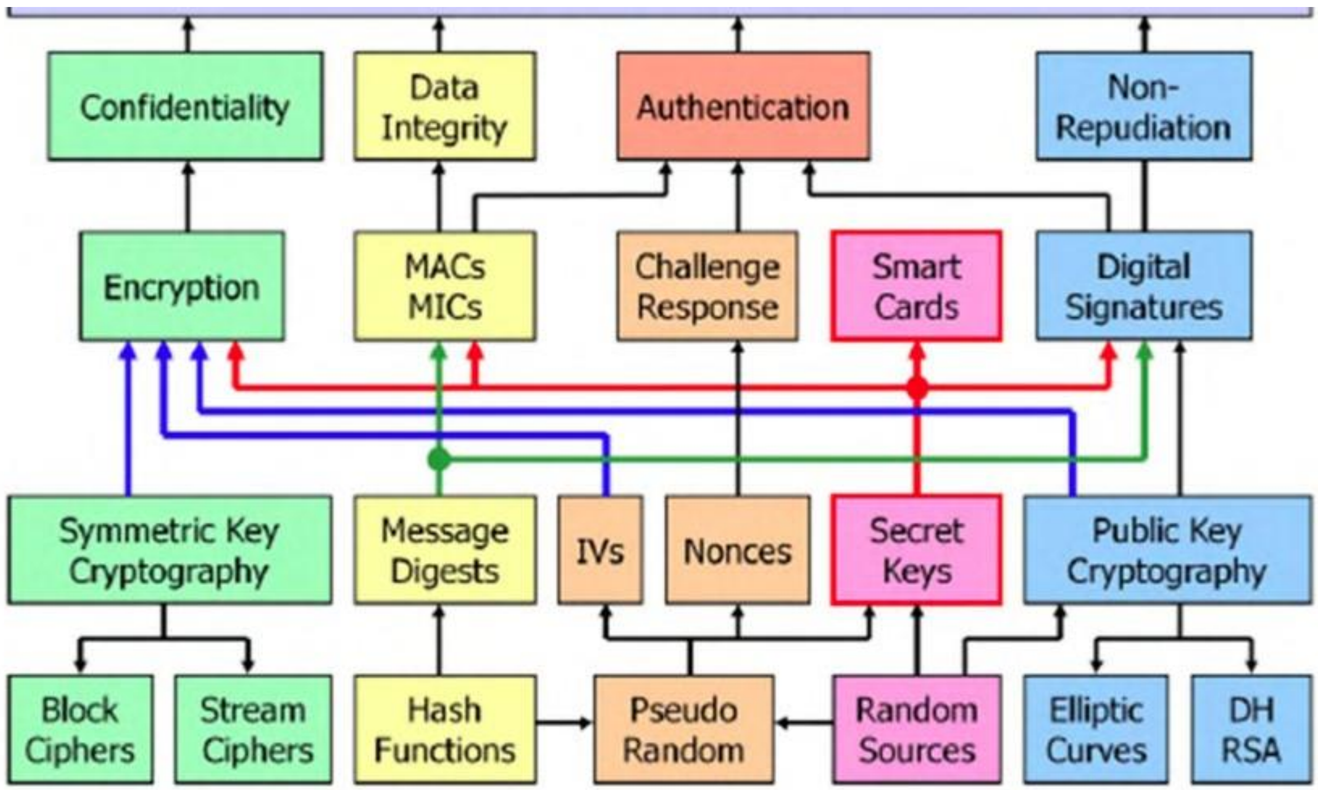
Which of the topics from the contents slide are you most interested in?

CIA++

FHV

- Confidentiality
  Valuable information or sensitive data must be protected from unauthorized access.
- Integrity
  Data must be protected from getting accidentally or mischievously changed either in its storage location or during transmission.
- Availability
  In mission critical environments the server and communications infrastructure must [usually] be available on a 24/7 basis.
- Authentication
  In any electronic transaction the true identity of the communication partners should be established.
- Non-Repudiation
  There should be a provable association between an electronic transaction and the person who initiated it.

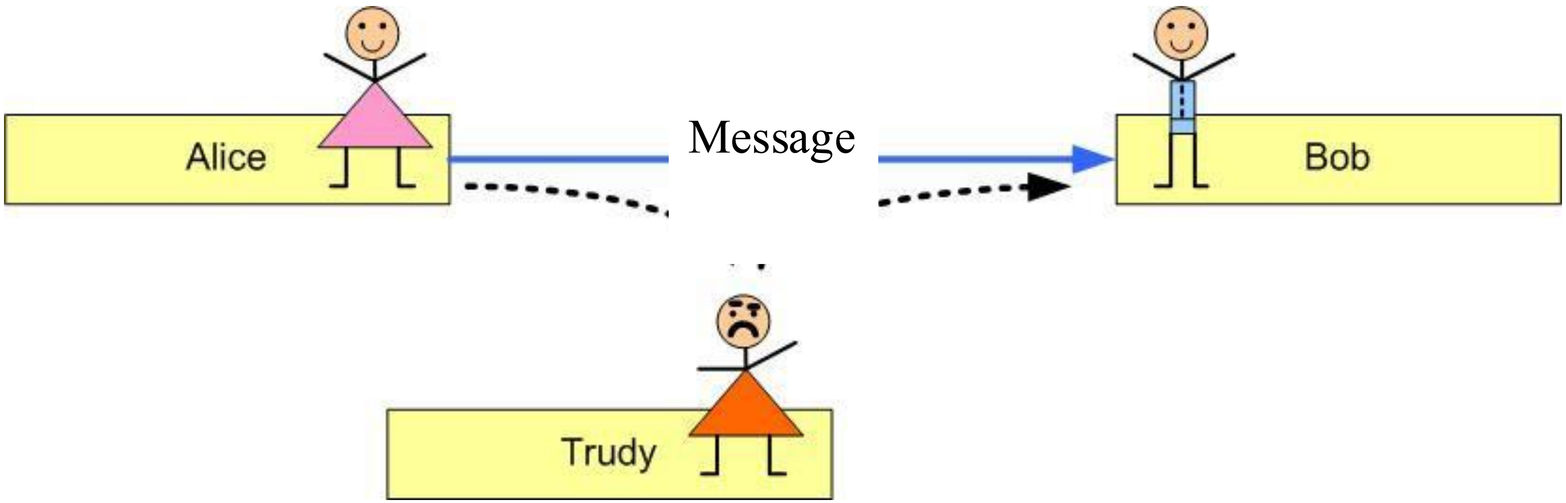
Building Blocks



Glossary:

DH	Diffie-Hellman public key crypto system
RSA	Rivest-Shamir-Adleman public key crypto system
IV	Initialization Vector, required to initialize symmetric encryption algorithms
Nonce	Randomnumber, used in challenge-response protocols

- Alice wants to **confidentially** communicate with Bob
- Trudy wants to wiretap the communication
- Alice and Bob encrypt the message



Cryptology
Basic Terminology

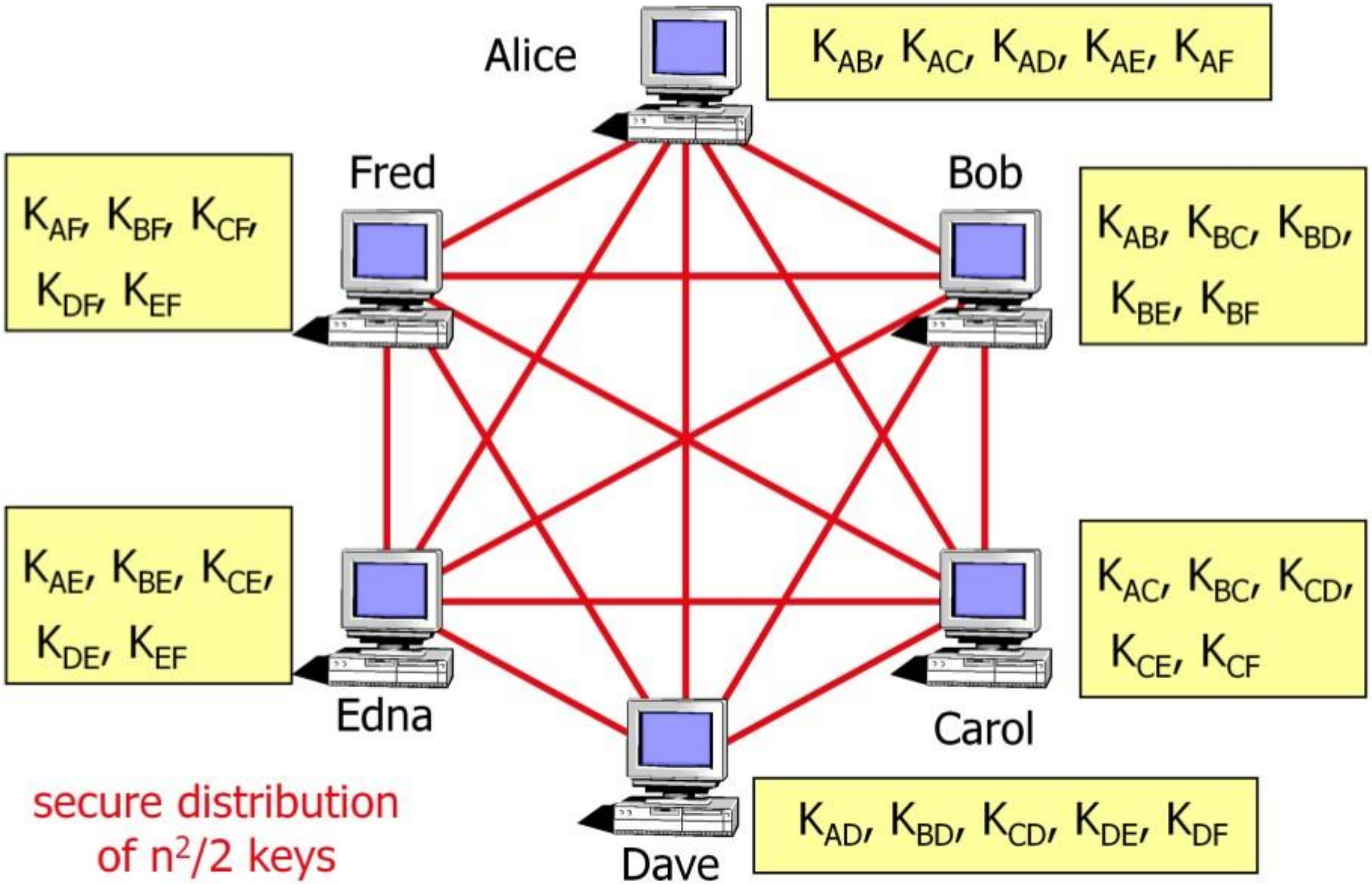
- *plaintext*
- *ciphertext*
- *key*
- *cipher*: cryptographic algorithm
  - (mathematical) function used for encryption and decryption.
- C=enc(K,P); P=dec(K, C) (encrypt, decrypt)

Cryptology

symmetric algorithms

- same key for encryption and decryption
- Problem:
  - sender and receiver must exchange the key in advance

- The Key Distribution Problem



- *Public Key Cryptosystems*
- encryption and decryption with different keys
  - Each communication partner has a secret (**private**) key for decryption
  - Using the **public key of the receiver** the sender can **encrypt** the plaintext;
  - this **public** key can be published (to everybody!), since it is impossible to derive the private key from the public.

C=enc(K_pub,P);;

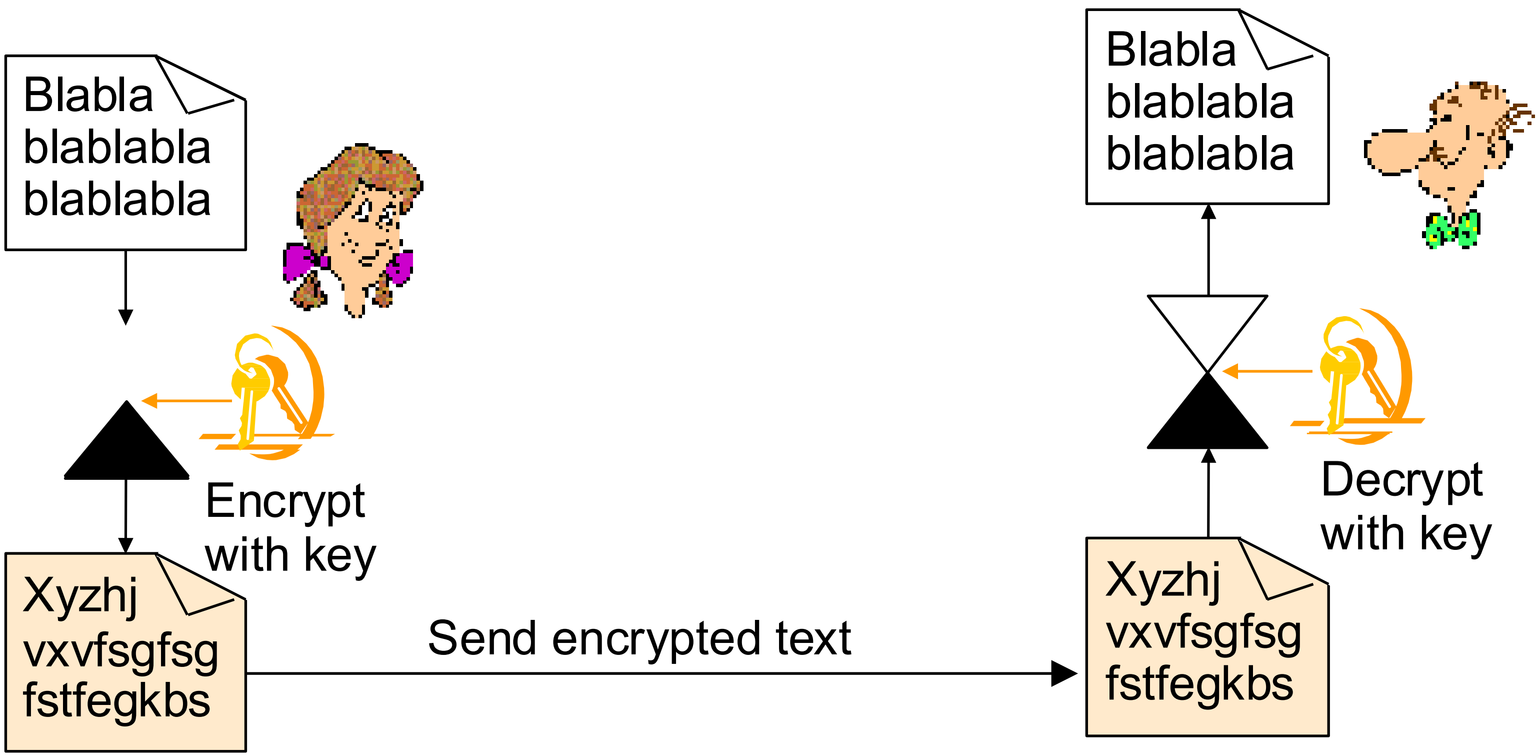
P=dec(K_priv,C)

Cryptology
asymmetric algorithms

- **Signature** of a message: “opposite direction”: Generation of the signature using private key of “sender” (=signer) and verification using public key

symmetric encryption

- Both have the **same** Key
- Must be exchanged in advance



asymmetric encryption for confidentiality

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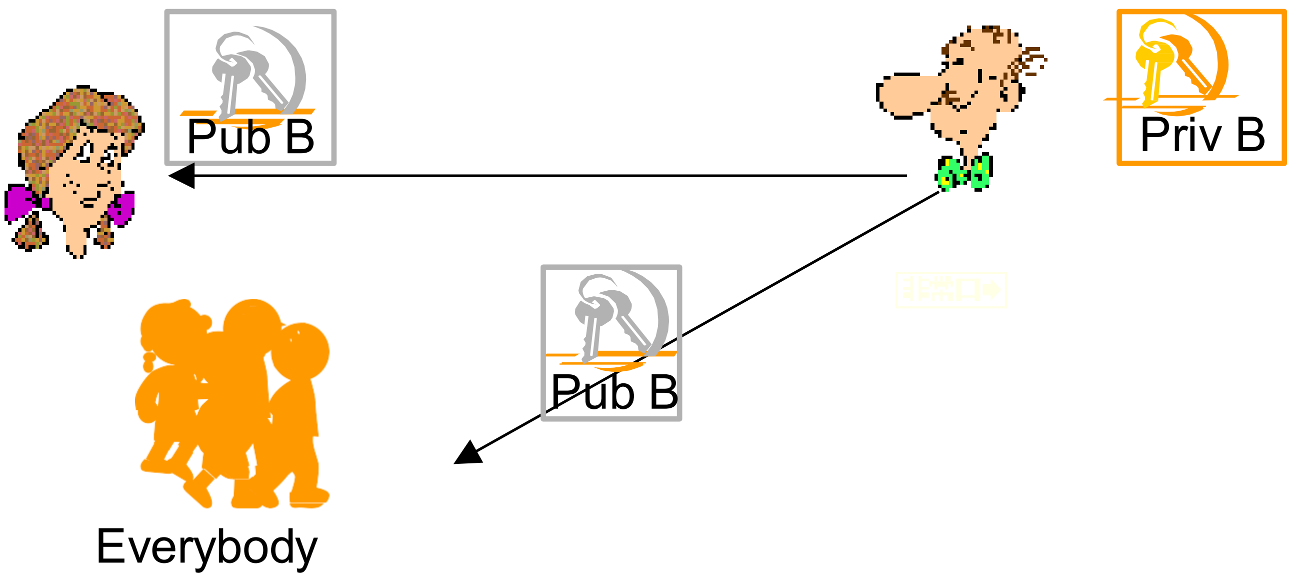
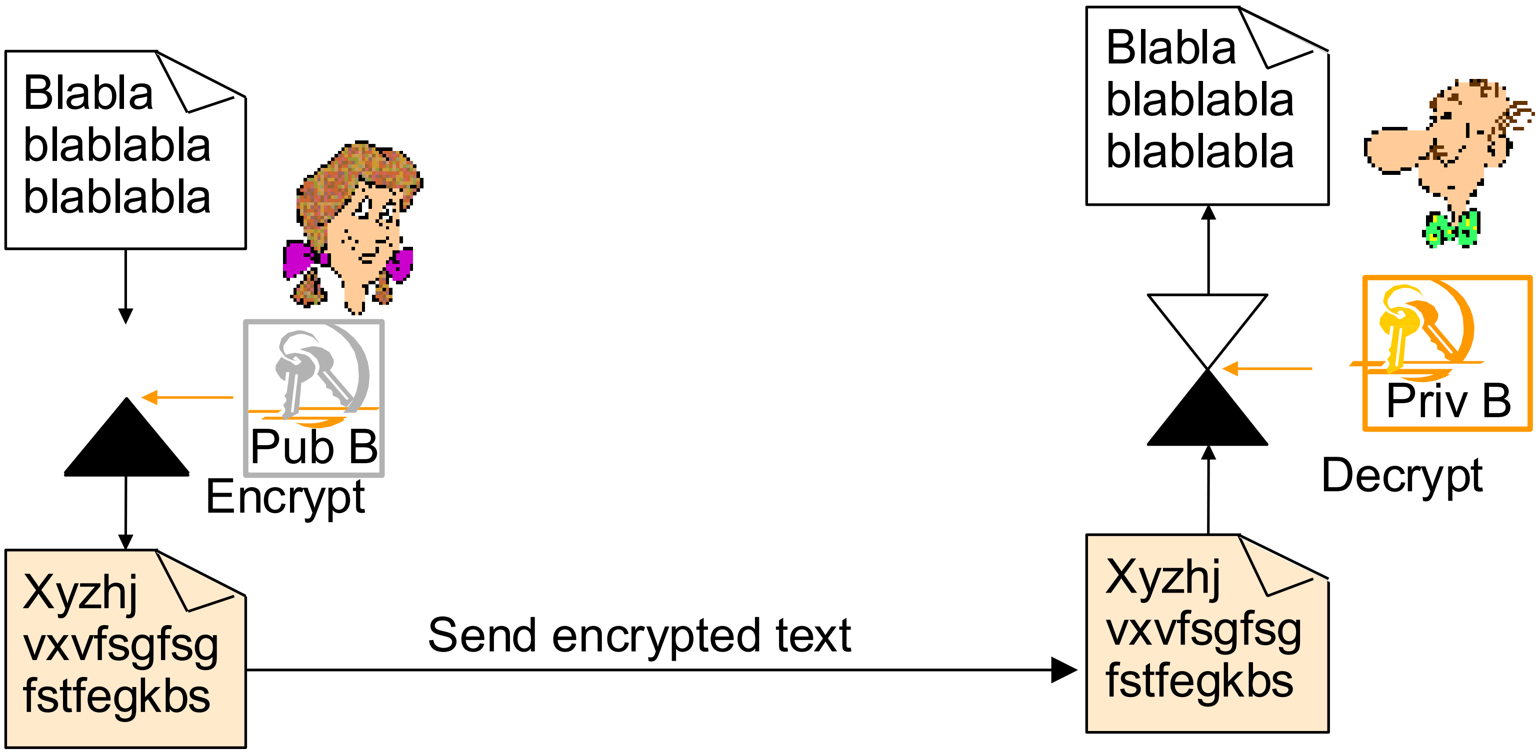
- Private and Public Key
- „In Advance“/ Before the encryption process:
  - Bob generates Private and Public Key
  - Stores Private Key on a secret place
  - Publishes Public Key (to everybody)



asymmetric encryption for confidentiality

- **Alice** can encrypt messages using **Public** Key of **Bob**
- Only Bob can decrypt message!



Authentication with asymmetric Key

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- **Bob** can authenticate himself using **his own Private Key** to other persons, since he is the only one knowing the Private Key!
- Alice verifies the identity using **Bobs Public Key**